EXECUTIVE SUMMARY

Predictive Forecasting of Care Load & Placement Demand

Government Stakeholder Brief

Purpose

The project, Predictive Forecasting of Care Load & Placement Demand, proposes an interactive analytical dashboard to support government stakeholders in understanding care-load trends, anticipating short-term demand, and reviewing potential capacity pressures. It brings trend analysis, forecasting, flow comparison, and scenario-based early-warning indicators together in one dashboard.

Project Objectives

- Forecast the number of children in HHS care.
- Estimate the future imbalance between intake and exits.
- Predict short-term discharge demand.

Secondary objectives are to provide early warnings for healthcare planners, quantify forecast uncertainty, and compare statistical and machine-learning forecasting approaches.

Analytical Approach

The dashboard is developed using Python and Streamlit, with data processing and visualization supported by Pandas, NumPy, Matplotlib, and scikit-learn. It provides interactive trend visualization, rolling averages, short-term forecasts, approximate forecast ranges, comparison of simple forecasting baselines, capacity-threshold scenarios, and KPI summaries.

Expected Value for Government Stakeholders

1. Anticipate workload changes by reviewing projected care-load trends.
2. Assess flow imbalance by comparing inflows and exits over a selected period.
3. Support discharge planning with projected discharge volumes.
4. Review capacity scenarios and identify forecast periods approaching a defined threshold.
5. Evaluate forecast reliability through model errors and uncertainty indicators.

Forecasts are intended to support professional and administrative judgment; they are not intended to automate decisions concerning individual children or placements.

Key Performance Indicators

The dashboard organizes performance review around forecast accuracy, surge lead time, capacity exposure, forecast stability, and model robustness. These indicators require agreed definitions and validation before official reporting. Capacity exposure should be assessed against an approved threshold, while accuracy and robustness should be established through repeated evaluation using reliable historical data.

Current Status and Important Consideration

The current dashboard is a demonstration prototype using simulated data. It illustrates the analytical workflow and dashboard functionality, but its numerical outputs are not official HHS statistics and should not be used for operational decisions.

Before operational adoption, the project requires authorized and validated historical data; standardized definitions for intake, transfers, care load, discharge, and capacity; data-quality checks and reconciliation of stock and flow measures; repeated chronological model validation; calibration of forecast uncertainty; and stakeholder approval of KPI definitions and capacity thresholds.

Conclusion

The project establishes a practical foundation for a forecasting-based decision-support dashboard for care-load and placement-demand planning. With validated operational data, agreed performance measures, and robust model evaluation, the dashboard could support more timely planning, improve visibility into potential workload changes, and provide government stakeholders with a structured analytical resource for reviewing care demand and placement capacity.

Executive Takeaway

The dashboard provides a practical prototype for anticipating care-load and discharge demand, identifying potential capacity pressure, and supporting evidence-informed planning. Its operational value depends on integrating validated historical data, establishing approved KPIs and capacity thresholds, and rigorously validating forecast accuracy before use in government planning decisions.